Project Information

This document contains one research assignment and one project with two milestones, which together form the core of your assessment. The research project will guide you in exploring how modern AI tools are transforming programming, with a particular focus on C# development. The two milestones will take you through the process of planning and building a practical program, ensuring that you apply the concepts you have learned in a structured and meaningful way.

In total the research assignment and milestones are marked out of 110 marks:

- Research – 30 marks
- Milestone 1 – 35 marks
- Milestone 2 – 45 marks

Milestone 2 carries slightly more weight since it assesses both your ability to implement a working program and to present it clearly. Your performance across these components will demonstrate both your ability to investigate current trends in programming and your skill in applying C# fundamentals to solve real-world problems.

Generative AI may be used as a study aid for understanding concepts; however, it must not be used to generate or submit assessed work. All submissions must reflect academic integrity and be the student's own original work.

Research Assignment

You are tasked with creating a short report on ***how modern AI tools are transforming programming, with a particular focus on C# development.***

In your report, you must:

- Investigate current challenges in C# development.
- Research and describe 5 trending AI-powered tools, including their features and integration with C#.
- Highlight the benefits and practical use cases of these tools.
- Discuss limitations and ethical considerations.
- Conclude with a future outlook on the role of AI tools in C# development.
- Provide proper references to your sources and make sure that you include intext citations.

Note:

- Any submissions without proper references and in-text citations will not be considered for marking.
- If markers identify that the work is not your own, contains plagiarism, or shows any form of academic misconduct, the research will automatically receive a mark of zero.
- Late submissions will not be accepted for marking.

Mark Breakdown

Section	Marks
Introduction	3
Current Challenges in C#	5
Research five trending AI-powered tools, outlining their features, C# integration, and practical benefits.	10
Limitations & Ethics	4
Conclusion & Future Outlook	4
Document Structure	4
Total	**30**

Milestone 1: Planning the Program

You are tasked to plan and design a **Budget Tracker program** for a group of university students who share an apartment and struggle to keep track of their shared monthly expenses, such as rent, groceries, utilities, entertainment, and so on. To solve this problem, you will design a program that allows them to record income and expenses, categorise transactions, and generate summaries of their spending habits. This solution is not limited to these features, and you are encouraged to enhance the system by adding additional relevant functionalities to improve usability and effectiveness.

Requirements

- Clearly define the main features and functionalities the Budget Tracker must have, focusing on income and expense tracking, categorisation, and summary generation.
- Identify the key data elements such as transactions, categories, and totals, and describe how they will be organised using basic data structures.
- Describe simple user interactions for inputting and viewing data, such as menu-driven console input.
- Outline the logical flow of the program, including adding transactions, updating totals, and generating summaries, without involving complex database operations.
- Include diagrams or charts that support your design, such as flowcharts and class diagrams that reflect the taught concepts. Using a suitable tool such as draw.io.
- Explain the methods and control structures you plan to use, such as loops and conditionals, to implement the program logic.
- Describe user interaction scenarios in more detail (how they will engage with the program)

Final Submission Requirements

A written planning document in PDF format (5-10 pages or equivalent) which includes the following:

- Solution idea and requirements of the system (functional and non-functional requirements)
- A flowchart and one diagram (structure chart) that demonstrates the core functionalities of the application.
- Clear explanations regarding the data structures and methods.
- A user interaction section that explains how students will engage with the program (menus, prompts, feedback).

Mark Breakdown

Section	Marks
Suitable solution idea & requirements	4
Flowchart	7
Diagram (structure chart)	5
Methods & Control Logic	5
Data structures	4
User Interaction	7
Document Structure	3
Total	**35**

Milestone 2: Development and Implementation

Build upon your planning from Milestone 1 to develop a working Budget Tracker program. This milestone focuses on implementing the core functionalities, ensuring the program can accept user input, process transactions, categorise expenses, generate accurate summaries and additional features. You will also refine the user interaction areas and ensure the program runs smoothly with proper control flow and data management.

Requirements

- Implement the main features planned in Milestone 1, including income and expense recording, categorisation, and summary generation. You may refine or modify the Milestone 1 design to improve the solution.
- Develop user input handling and output display, using console elements.
- Write methods/functions that encapsulate key operations such as adding transactions, calculating totals, and generating reports.
- Use control structures to manage program flow.
- Utilise the data structures identified in Milestone 1 to store and manage transactions and categories.
- Refine the user interactions to ensure usability and clarity.
- Ensure the program runs correctly and is user-friendly.

Final Submission Requirements

- Write methods/functions that encapsulate key operations such as adding transactions, calculating totals, and generating reports.
- Use control structures to manage program flow.
- Utilise the data structures identified in Milestone 1 to store and manage transactions and categories.
- Program tested to ensure it works as expected and is usable by users.
- Clear and organized presentation of the program, demonstrating your understanding of the implementation.
- Ensure to upload all the necessary documents

Presentation Instructions

- You will be presenting both Milestones
- Prepare a PowerPoint presentation with at least 5 pages explaining the planning process that you underwent in Milestone 1 as well as the purpose of your program, how your system works and challenges faced.
- After the PowerPoint presentation you will also do a live illustration of your system while demonstrating your understanding of the code, coding and design concepts you used as a part of you mark breakdown. You will be assessed based on your understanding of the concepts, not just on having a working solution.

Mark Breakdown

Section	Marks
Appropriate use of data structures	5
Implementation of functional requirements aligned with the problem solution	10
Correct and effective use of control structures	5
Logical flow and structure of the application	5
Overall application usability	5
Additional features implemented	5
Presentation and project defence	7
Total	**45**

Important Information

- The Research assignment is part of Milestone 1 and is submitted as group work. The same groups are also used for Milestone 2. You will be automatically assigned to groups of 4 or 5 students.
- There will be only a single submission per group on BC Connect. (Unless communicated otherwise)
- Belgium Campus uses software that can scan for plagiarism, AI, and paraphrasing.
- Students caught doing this will get 0 for this assignment.
- Your lecturer may deduct marks if the rules, instructions, or guidelines for the project documentation are not followed.
- Late assignments will not be accepted. Missing the deadline or not presenting your project results in an automatic 0.
- If you have any documentation, please ensure it is properly formatted, as discussed during ACW.